Random Widget Works (RWW)

Random Widget Works is a manufacturer and wholesaler of widgets. RWW widgets are marketed to the nation's almost 10,000 businesses that incorporate widgets into their software and hardware.

Theirs is a well-known story – a small startup in the garage of a visionary. In 1994, It began with a husband-and-wife team working in their small suburban neighborhood on the outskirts of Atlantis. The couple worked day and night to develop the software widgets, part of a must-have firmware add-on to any good product. After two years of scraping by, the value of their product was recognized by one of the giants – Hierarchical Access Limited, Inc. The corporate giant set up the pair as a subsidiary – Random Widget Works, Ltd. – in a 12,000 square foot manufacturing plant and provided the leadership, resources and support needed to grow the small start-up to a Fortune 1000 small-to-medium (SMB) business within 10 years.

THE RWW FACILITIES

Today, the husband and wife team of Ralph and Alice Winston serve as the Chief Operations Officer and Executive Assistant to the COO, respectively. The original facility was eventually outgrown and a much larger 50,000 square foot facility and adjacent office complex was leased. The manufacturing facility produces thousands of widgets per day. The 15,000 square foot office building houses corporate offices, including sales, operations, and IT.

MANAGEMENT TEAM

CEO Winston Ryan brings his decades of experience in manufacturing from the automotive industry to the organization. Mr. Ryan was hired by RWW after the corporation purchased the rights to develop the "Winston Widget" (the original name of the company). CFO, Nishandra Jackson, was hired based on his extensive experience as an accounting executive with expertise in the pharmaceutical manufacturing and sales industry. In 2000, Leslie West was hired as the company's Chief Marketing Officer. Finally, Regina Nash serves as the first Chief Information Officer, hired in 2010. The company's current organization chart is provided at the end of this document.

RWW COMPANY DESCRIPTION

THE ENVIRONMENT

RWW is an Atlantis-based company that will provide firmware widget solutions for a host of developmental needs. The company manufactures and sells widgets wholesale to developers in 22 states in the U.S. and 14 countries overseas.

RWW'S MISSION, VISION, and VALUES Statements

Mission

Random Widget Works designs and manufactures quality widgets and associated equipment and supplies for use in modern business environments.

Vision

Random Widget Works will be the preferred manufacturer of choice for every business's widget needs, with an RWW widget in every gizmo in use.

Values

Random Widget Works values commitment, honesty, integrity, and social responsibility among its employees and is committed to providing its services in harmony with its corporate, social, legal, and natural environments.

RWW Personnel

KEY EMPLOYEES

- Winston Ryan, CEO: Prior to RWW, Winston worked as a database developer. Mr. Ryan completed his MBA at the University of Atlantis with a concentration in IS in 1989. He received his BBA in Entrepreneurship in 1986 from the University of Yoknapatawpha.
- Leslie West, CMO: Ms. West has worked at Mega Care, one of the nation's largest corporate manufacturing providers. She also provided technical support at WizBank, a manufacturer of finance software. Ms. West completed her MBA at the University of Atlantis in 1995 with an emphasis in marketing and operations management.
- Nishandra Jackson, CFO: Ms. Jackson supports the RWW management team in managing its financial infrastructure as well as capital development. She also oversees the financial administrator, who will manage financial structure, cash position, financial statement reporting and benefits administration. Ms. Jackson completed her MBA at the University of Atlantis in 1995 with an emphasis in finance and accounting.
- Regina Nash, CIO: A graduate of the Wheaton School of IT with a Masters of Science in Information Technology, Ms. Nash oversees all technical infrastructure, software development and customer technical support.

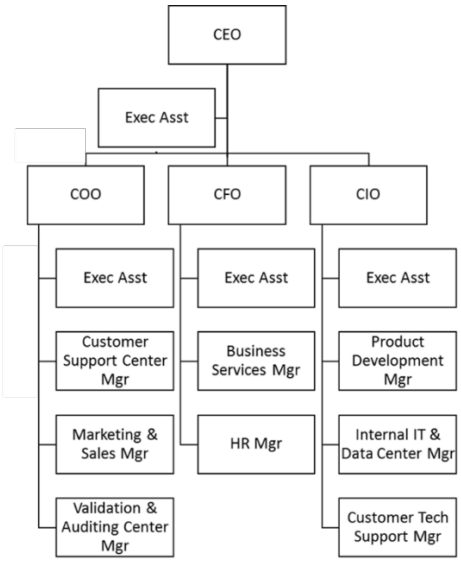
BOARD OF DIRECTORS

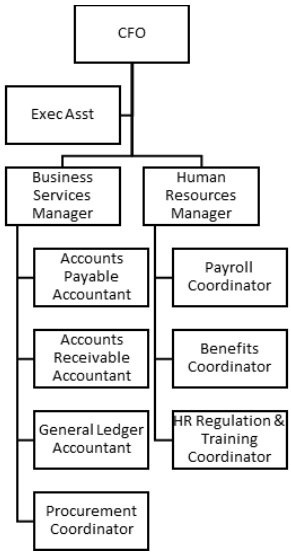
The positions of chairman of the board and vice chairman have been reserved for RWW's initial investors. The company's attorney also serves on the board.

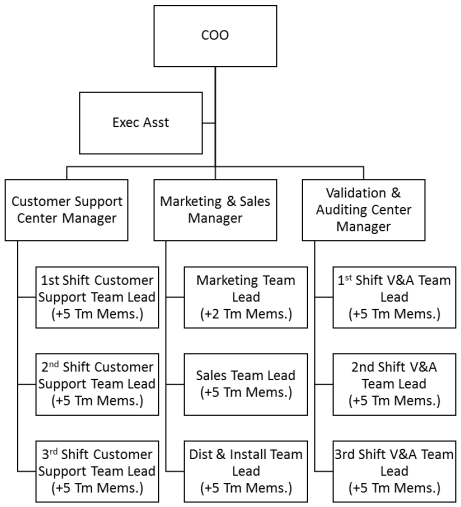
MANAGEMENT STRUCTURE

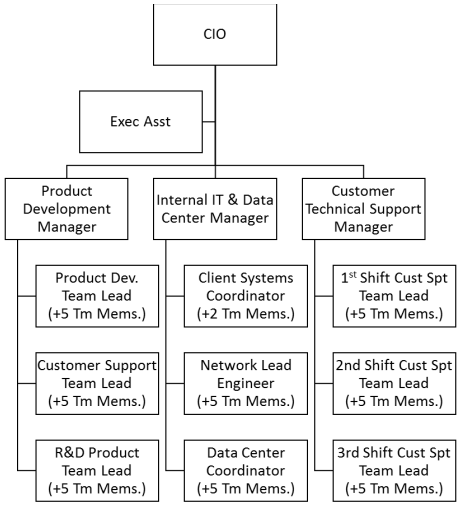
Corporate officers are involved in the day-to-day activities of RWW. They directly oversee the software development team and product support team. Business Development, Finance, and Operations report to the officers but operate with a high degree of autonomy and flexibility. Advisors to the firm direct regulatory research, product functionality and usability testing. Management responsibilities at RWW are shown on the following organizational charts.

Note: for this project, don't worry about employee names. When needed, use their position for roles & responsibilities.









Mgr = Manager

Tm Mems = Team members

RWW IT Operations & Infrastructure

There currently is no formal cybersecurity department or structure at RWW. For such an informal business, RWW has a very formal, mature IT department, with its centralized data center on the second floor of the corporate office building. RWW's IT department manages its data center and internal networks and applications. All corporate applications have a secure Web interface that employees use to connect to the systems (no local data stored on PCs). Most of the server-based applications have external SQL databases stored in an encrypted format on the same physical server. There are a few specialized applications with internal data storage, also encrypted. The Corporate offices connect to the Internet through state-of-the-art firewalls through two different ISPs providing redundancy. In the data center, all networked applications each have their own dedicated server, with all data/databases backed up daily to cloud–based storage provided by RWW's cloud services division – Redundant Redundancies, Ltd. (a.k.a. Software-as-a-Service). All applications, including those with internal data, are backed up weekly. RWW's data center meets the high-quality standards of the industry and provides systems that assure redundant power availability, temperature control, and fire suppression.

> *For many companies, the level of security, availability, disaster recovery, and back-up, provided by a software-as-a-service provider far exceeds that which they can provide themselves. RR Ltd. provides multiple layers of data protection to ensure RWW's business and business associates' data protection. It manages thousands of petabytes of data for customers all over the world, with multiple off-site remote backups around the globe. Your data is safe with RR Ltd.*

All employees access their networked applications over desktops (no laptops, tablets or mobile devices are allowed to connect to the corporate network) through physical LAN connections within the branch offices, which uses a secure VPN connection for all traffic between the corporate networks and client systems. RWW does not allow anyone *except* Corporate executive and senior IT managers to access any of its resources remotely (through Corporate VPN connections) and does NOT allow Wi-Fi networking in any of its offices. RWW Executives have company-issued wireless smart phones with VPN connections to access corporate data or networks on those devices. The current *informal* policy is no BYOD on premises. All employees must keep

their cell phones in their vehicles, briefcases, or purses and only use them during authorized breaks in designated areas.

The RWW Secured Server Data Center contains the following:

Server Rack 1: (Financial & Planning Systems)
Rack 1 Servers containing most of the Corporate IT Applications: The following Corporate IT applications are each installed on their own server. These include (not are not limited to):

Server 1A: Human Resources Management System (HRMS) and HRMS SQL DB
-Includes personnel files, employee recruitment, benefits, pay, and evaluations.

Server 1B: Enterprise Planning System (EPS) and Planning SQL DB
-Includes files to support company strategic planning and resource management.

Server 1C: Financial Management System (FMS) and FMS SQL DB
-Includes accounting systems like general ledger, payroll, accounts payable, and accounts receivable.

Server 1D: Project Management and Projection Systems (PMPS) and PMPS SQL DB
-Includes project management planning support.

Server 1E: Enterprise Time and Labor System (ETLS) and ETLS SQL DB
-Managerial time management application including salaried and hourly employee time tracking, paid time off (PTO), etc.

Server Rack 2: (Sales and Support Systems)

Rack 2 Servers containing most of the Corporate Sales and Service Applications: The following applications are each installed on their own server. Some of these have **internal** databases/data stores. These include (not are not limited to):

Server 2A: Enterprise Sales System (ESS) and ESS SQL DB
-Includes sales support, product orders, invoicing, procurement, sales contracts, and related documents for customers and suppliers.

Server 2B: Supplier Relationship System (SRS) and SRS SQL DB
-Specialized software to support tracking and relationship management for supplier products and services.

Server 2C: Microsoft Office 365 Web Applications (MS Office) and Employee Files OneDrive DB
-MS Office applications (Word, PowerPoint, Excel, and Visio).
All employee documents/files are stored on the Microsoft cloud OneDrive server, not on local systems. Any files edited on local systems are synchronized in real time and wiped at the end of each business day (on employee logout).

Server 2D: Helpdesk Application (HDA) and HDA SQL DB
-A specialized application that tracks external customer issues related to orders, shipping, and product receipt. Also manages internal helpdesk support providing user trouble tickets related to authorized company computer system use.

Server 3:
Rack 3 Servers containing most of the public-facing Internet applications: The following applications are each installed on their own server. Some of these have **internal** databases/data stores:

Server 3A: Outlook Email Server and Outlook Email DB
-MS Outlook 365 (Web) application with proprietary email storage. All stored email in employees' *inbox* and *sent* mail folders are purged every 90 days, unless moved to internal "archive" folders. All employees' *junk* and *deleted* mail folders are purged every 30 days.

Server 3B: RWW Corporate Web Server (Web) with Web pages (no DB)
-All public-facing websites used by the company. There is no ecommerce site, only static web pages, with some web forms to forward potential customer and employee information to the appropriate department.

Server 3C: Primary DNS Server (PDNS) and internal DNS database
-provides domain name services for the organization
(note: secondary DNS server support is provided by parent company on its servers and is thus outside the scope of this project).

Server 3D: Windows Active Directory Server (ADS) and internal ADS data set
-AD is used to manage login authentication, employee access to other resources (e.g., applications and services).

Server 3E: VPN remote access system (VPN)
-Application that allows corporate executives to access, and IT personnel to remotely manage all resources. Has no internal data, authenticates with ADS instead.

Note: All SQL databases are backed up daily to the RR Ltd. Cloud. All applications (including those with internal data) are backed up weekly to the RR Ltd. Cloud. Microsoft manages its own Office/OneDrive cloud systems and data. *All off-site backups are outside the scope of this project.*